



SIMULATING ALCOHOL-LIKE VISUAL IMPAIRMENTS IMPAIRMENTS IN VIRTUAL REALITY

Effects on Spatial Working Memory

XR Prototype – Task 1 of a 4-Task Cognitive Framework

6741626, 6682848

Research Motivation

Why study alcohol-like impairment in VR?



Alcohol affects memory, perception, and coordination



Real-world testing is unsafe and hard to control



VR allows safe, repeatable, precise simulations



Hafstrom et al. (2014) research into effects of alcohol on stability using VR but not through using an embodied task.

Why it matters:



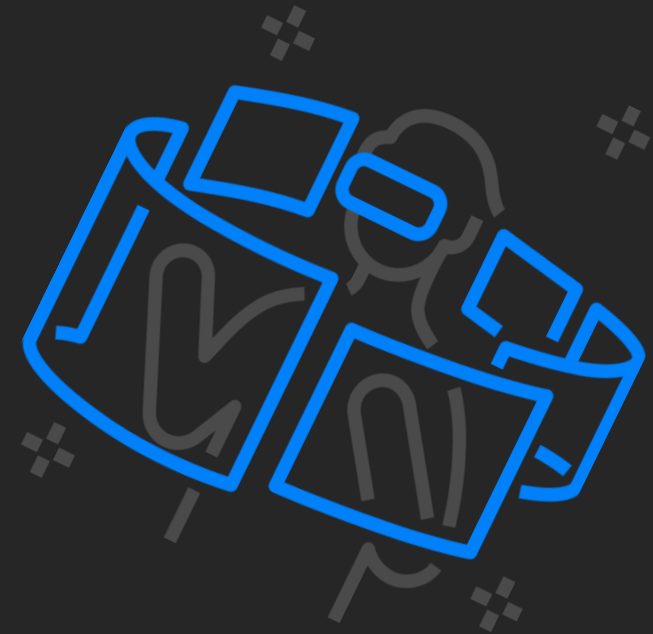
Accidents & falls



Driving & decision errors



Everyday spatial tasks



Research Questions



Primary Question: How does alcohol-like visual impairment affect spatial working memory during a path recall task in VR?



Navigation Errors

Does impairment increase navigation errors?



Path Accuracy

Does impairment reduce accuracy on harder paths?



Completion Time

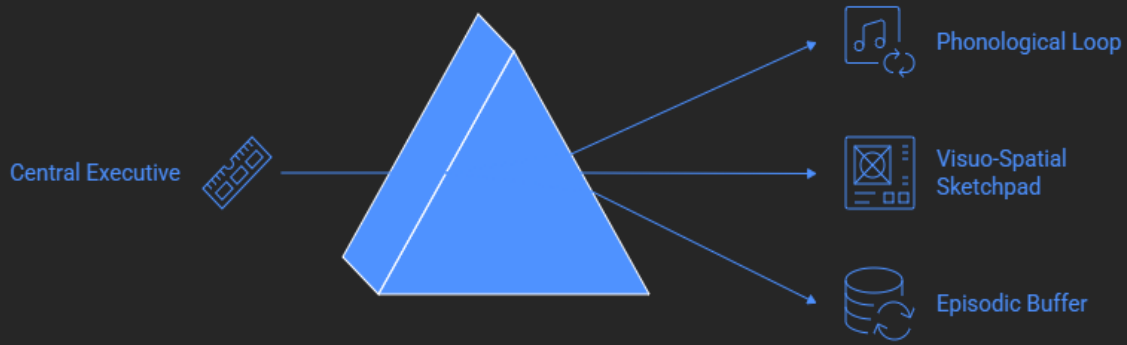
Does impairment increase completion time?

H1: Alcohol-like visual impairment will increase navigation errors.

H2: Impairment will decrease path recall accuracy, particularly in relation to path difficulty.

H3: Impairment will increase completion time.

Psychological Foundation



🧠 Baddeley & Hitch's Model

Visuospatial Working Memory

The Visuospatial sketchpad is responsible for spatial memory, visual memory, and spatial navigation

Two Cognitive Phases:

- 1 Encoding the spatial path**
Initial perception and storage of spatial information
- 2 Recall and execution from memory**
Retrieval and application of stored spatial information

🔑 Alcohol's Impact

- Spatial encoding & retention
- Motor planning and coordination
- Attentional stability

🧠 Relevant Brain Systems



Hippocampus
Spatial memory



Prefrontal cortex
Working memory,
decision-making



Cerebellum
Motor control

The bigger picture: A 4-Task Cognitive Framework

Our VR environment is one component of a broader system exploring alcohol-like visual impairment across multiple cognitive abilities.



Spatial Working Memory

Player navigation and memory component.



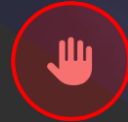
Sustained Attention / Vigilance

Monitoring and focus maintenance over time.



Risky Decision-Making

Judgement calls and risk assessment under impairment.



Motor Coordination & Fine Motor Skills

Precision and control movements under impairment.



Goal:

Simulate how alcohol-like perceptual distortions affect different cognitive systems in a safe and controlled VR environment.

Core Concept Overview

The Core Idea:



Spatial navigation memory task



Path recall from memory



Performance measurement

5x5 Grid Environment

0,0	0,1	0,2	0,3	0,4
1,0	1,1	1,2	1,3	1,4
2,0	2,1	2,2	2,3	2,4
3,0	3,1	3,2	3,3	3,4
4,0	4,1	4,2	4,3	4,4

Start Path End

Task Details:

Two difficulty levels:



- Easy path
- Hard path

Two conditions:



- Sober vision
- Alcohol-like impaired vision

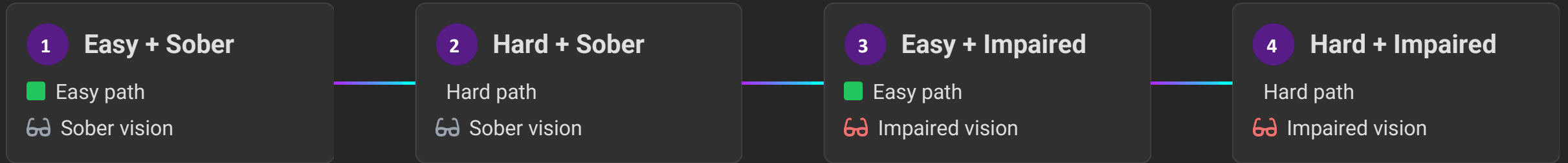
Measures:



- Accuracy
- Errors
- Completion time

User Flow and Visual impairment

Participants complete all 4 counterbalanced combinations of difficulty levels with sober/impaired vision conditions.



Blurred Vision

- ✓ Simulates reduced acuity under alcohol
- ✓ Strong contrast between sober vs impaired

i Implemented using post-processing effects



Camera Shake

- ✓ Mimics instability & impaired motor control
- ✓ Safe for VR (avoids disorientation triggers)

i Implemented using camera rotation vectors



Key benefit: Strong contrast between sober vs impaired conditions for clear measurement

Why This Approach Works



Psychologically Grounded

Built directly on visuospatial working memory principles



Quantifiable Behaviour

Errors, correct tiles, completion time, difficulty ratings



Embodied VR Interaction

Physically walking the path increases ecological validity



Feasible for Scientific Testing

Easy to counterbalance; consistent across participants.



Simple & Controlled

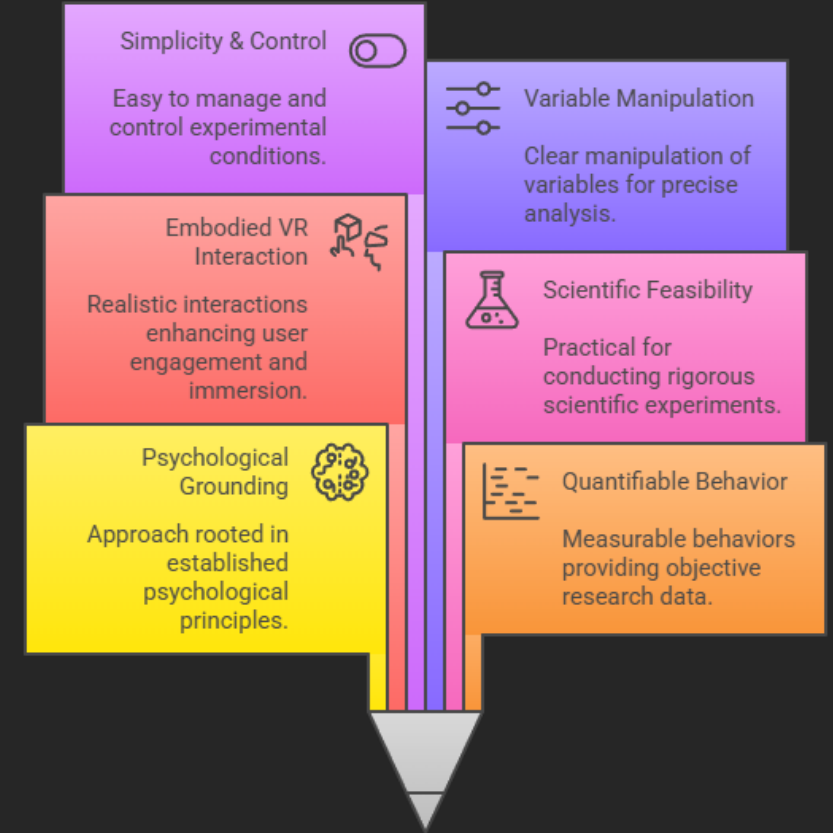
5x5 grid, fixed sequences, replicable experience



Clear Variable Manipulation

Sober vs impaired vision can be precisely controlled

Building Blocks of VR Research



Calibration Room Design



Key Design Elements



Living-room style

Reduces cognitive load & prevents distraction



Mini practice grid

Practice movement & tile interaction



1:1 real-world scaling

VR room matches physical dimensions



Spatial orientation

Clear landmarks for navigation

Benefits of Calibration Room

✓ Helps users feel grounded and safe

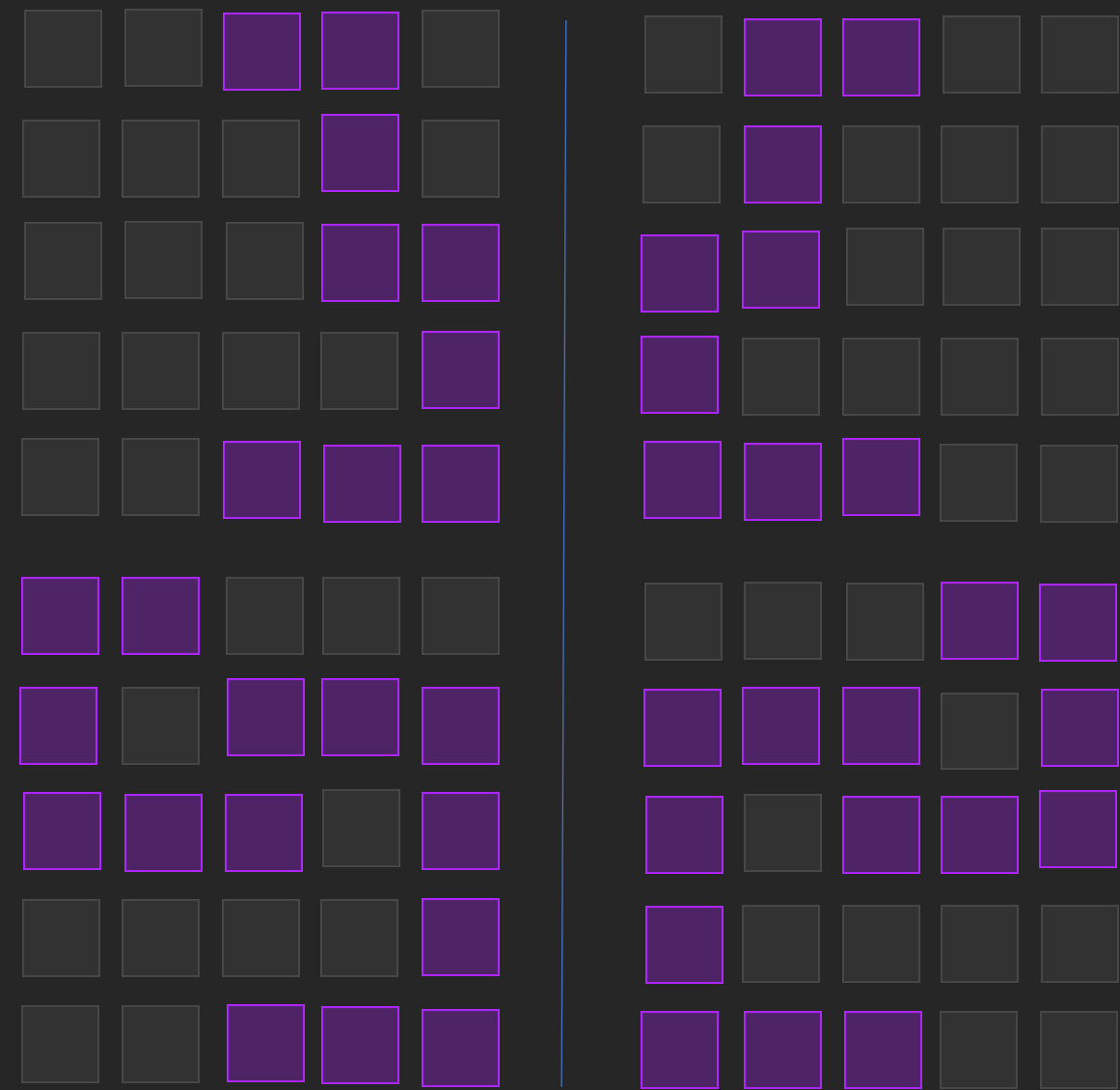
✓ Mirrors real space with virtual markers

✓ Enhances spatial orientation & embodiment

✓ Minimizes sensory mismatch to reduce VR sickness

VR Environment Architecture

5×5 Grid Layout



Grid Features:

- ✓ • Clear, controlled spatial structure
- ✓ • Supports quantifiable path design
- ✓ • 25 interactive tiles

Environment Design



Minimalist Architecture

Uncluttered environment reduces visual noise to focus attention on the task



Tile Interaction System

Tiles change colour when stepped on with upward-facing light for feedback



Lighting Design

Smooth lighting transitions to reduce visual discomfort and VR sickness



Walking Comfort

Spacious layout designed to reduce need to look down at tiles

Performance Metrics and Measurement

Behavioral Measures



Correct Tiles

Measures accuracy in path following



Off Path Steps

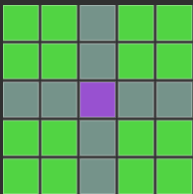
Counts deviation from intended path



Completion Time

Records task completion in seconds

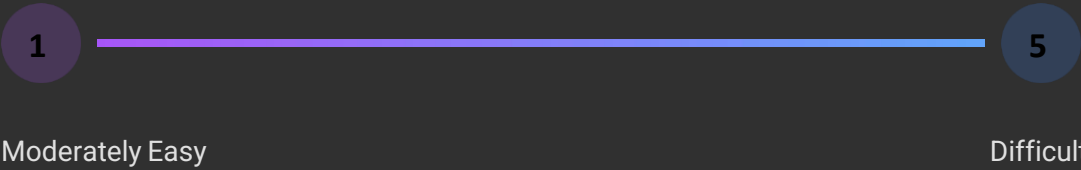
Error Heatmap



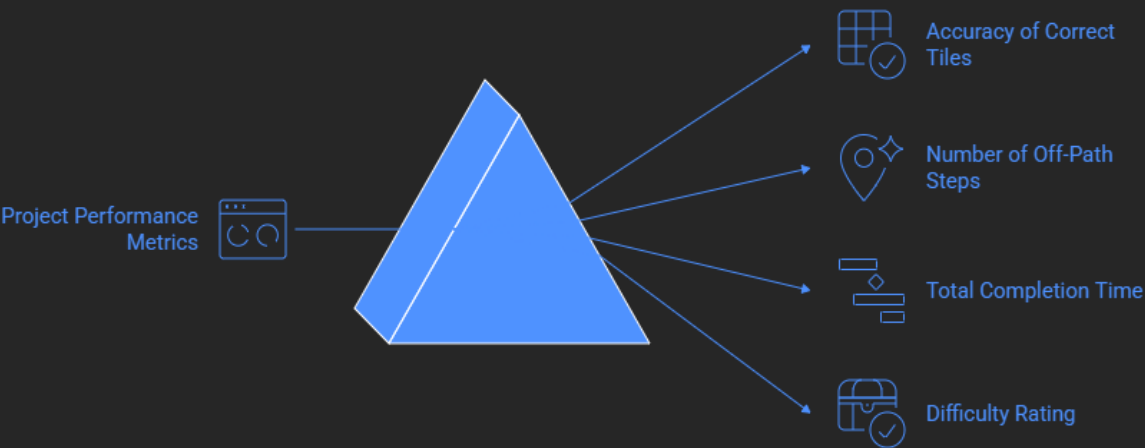
Visual representation of error frequency across the 5×5 grid

Subjective Measures

Difficulty Rating



Participants provide qualitative feedback on task difficulty



Walkthrough and explanation



Walkthrough Video

- This video shows a walkthrough of the easy path with visual impairment implemented
- Demonstrates impaired navigation and visual distortion
- Shows the path participants must recall

Applied Visual Impairment Effects

- Introduced blur and distortion using a post-processing material in Unreal Engine
- Applied a camera shake blueprint to simulate instability and reduced motor control
- Combined blur + shake for a realistic alcohol-like visual impairment
- Effects were attached directly to the player camera, so they remain active during navigation
- Intensity calibrated to ensure paths were challenging but still completable

Current Limitations



No Full Body Tracking

Limited to solely hand tracking



Restricted Room Size

Constrained physical space limits natural movement



Single Environment Type

Currently only tests one type of environment.



Visual-Only Impairment

Does not simulate other alcohol-induced effects

Single Environment Type



Full Body Tracking



Visual-Only Impairment



Restricted Room Size



Research Conclusions and future direction



VR Spatial Memory Task

Successfully developed with sober vs impaired conditions



Theoretical Grounding

Built on visuospatial working memory theory



Visual Effects Implementation

Alcohol-like effects safely & consistently implemented



Controlled Design

Counterbalanced, controlled approach suitable for research



Performance Metrics

Clear, quantifiable measures of performance



Framework Integration

Forms Task 1 of a broader 4-task cognitive framework



Multiple Impairment Levels

Graduated severity scales to simulate varying degrees of alcohol impairment



Dual-Task Layers

Add cognitive distractions to simulate divided attention scenarios



Full Body Tracking

Implement foot and full body tracking for more natural interaction



Expanded Environments

Create diverse spatial contexts beyond the current grid layout



Advanced Analytics

Implement machine learning for deeper performance insights

References

Academic References

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